

WASAGAMING, MB, Canada

WMO: 714440

Lat: **50.6550N**Lon: **99.9419W**Elev: **2058**StdP: **13.64**Time Zone: **-6.00 (NAC)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-32.4	-27.0	-39.4	0.6	-31.9	-33.4	0.9	-26.5	14.0	9.9	12.3	8.8	0.7	40	0.627

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	22.7	82.6	66.2	79.3	64.6	76.4	62.6	69.6	78.1	67.2	75.5	65.2	73.2	7.0	220

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
66.8	106.6	73.9	64.3	97.5	71.3	62.0	89.8	69.4	35.0	78.1	33.0	75.5	31.3	73.2	81.9

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 14.0	12.2	11.0	DB	-39.6	88.6	5.5	3.4	-43.5	91.0	-46.7	93.0	-49.8	94.9	-53.8	97.4	(4)
(5)			WB	-39.5	73.6	4.9	3.4	-43.0	76.0	-45.9	78.0	-48.6	79.9	-52.2	82.3	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	33.5	1.4	5.0	17.9	34.0	46.6	56.9	62.2	60.4	51.2	37.5	20.5	6.4		
	DBStd	24.15	14.89	14.19	14.45	11.20	8.53	6.97	5.78	6.11	8.03	8.66	12.03	13.80		
	HDD50	7147	1506	1260	996	485	167	17	0	2	80	397	885	1352		
	HDD65	11583	1971	1680	1460	930	570	256	124	169	419	852	1335	1817		
	CDD50	1120	0	0	0	6	63	224	379	323	115	10	0	0		
	CDD65	82	0	0	0	0	1	13	38	26	4	0	0	0		
	CDH74	1192	0	0	0	1	54	177	474	396	87	3	0	0		
	CDH80	227	0	0	0	0	9	29	85	94	10	0	0	0		
(14) Wind	WSAvg	4.6	3.8	4.1	4.7	5.4	5.6	4.9	4.5	4.4	4.6	5.0	4.7	3.7		
(15) Precipitation	PrecAvg	21.7	0.9	0.7	1.5	1.2	2.9	3.8	2.8	2.6	1.9	1.8	0.9	0.9		
	PrecMax	27.2	1.4	1.5	5.2	2.4	6.3	6.5	6.4	6.6	3.3	5.4	3.5	2.0		
	PrecMin	16.7	0.4	0.0	0.4	0.4	0.8	1.5	1.1	0.5	0.2	0.5	0.1	0.2		
	PrecStd	2.8	0.3	0.4	1.2	0.7	1.4	1.5	1.4	1.5	0.8	1.1	0.9	0.5		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	37.0	40.1	54.4	69.9	80.2	83.6	86.1	87.7	81.2	71.0	53.4	39.1		
		MCWB	32.9	34.5	45.2	51.8	59.5	66.7	70.2	67.4	62.8	54.4	45.9	34.5		
	2%	DB	32.0	34.8	46.8	62.9	74.0	78.7	82.4	82.5	76.4	62.5	45.8	33.4		
		MCWB	29.8	31.1	39.2	47.2	55.0	63.3	67.2	66.3	60.9	50.0	39.1	30.8		
	5%	DB	28.0	30.6	41.9	58.1	69.0	75.2	79.2	78.4	71.9	56.8	40.3	28.8		
		MCWB	26.7	28.5	36.1	45.1	52.6	61.1	65.9	64.5	59.2	47.2	35.5	27.0		
	10%	DB	23.4	26.5	36.8	52.9	64.2	71.7	76.1	74.8	67.0	51.9	35.9	25.3		
		MCWB	22.4	24.9	33.1	42.3	50.3	59.5	64.0	62.6	56.4	44.7	32.9	24.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	33.6	35.4	44.7	52.9	62.1	71.5	74.0	72.3	67.5	55.6	45.6	34.7		
		MCDB	35.9	38.6	51.7	65.8	72.7	80.6	81.0	81.0	74.9	65.9	53.5	38.2		
	2%	WB	30.1	31.5	39.8	49.1	57.7	66.4	70.0	68.6	63.7	52.0	39.9	30.7		
		MCDB	32.1	33.9	46.0	60.6	68.1	74.3	78.3	78.0	72.1	60.5	44.8	32.9		
	5%	WB	26.6	28.3	36.3	45.9	54.8	63.5	67.7	66.6	60.8	48.5	36.0	27.1		
		MCDB	27.9	30.5	41.1	56.0	65.6	71.3	75.8	75.1	69.1	55.0	40.0	28.6		
	10%	WB	22.6	24.7	33.5	43.1	52.0	61.2	65.6	64.5	57.6	45.4	33.0	24.3		
		MCDB	23.5	26.4	36.7	52.6	61.5	68.7	73.6	72.4	65.3	51.3	35.6	25.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	21.1	22.9	22.8	22.7	23.7	21.6	22.7	23.7	22.8	18.0	16.5	19.1		
		MCDBR	23.3	23.6	25.3	30.7	32.9	27.1	27.5	30.3	31.5	29.2	23.3	22.2		
		MCWBR	21.1	19.9	18.0	17.5	16.6	14.2	14.7	16.1	17.5	17.3	16.6	19.6		
		MCDBR	22.7	22.0	23.0	28.1	26.9	22.5	23.8	24.9	26.4	25.7	21.7	21.2		
	5% WB	MCWBR	20.7	18.7	16.9	16.8	15.2	13.2	14.3	14.8	16.2	15.9	16.0	18.9		
(40) Clear-Sky Solar Irradiance	taub		0.250	0.252	0.298	0.328	0.341	0.354	0.369	0.366	0.333	0.282	0.253	0.227		
	taud		2.280	2.305	2.289	2.396	2.419	2.406	2.379	2.383	2.462	2.574	2.479	2.373		
	Ebn at Noon		252	285	288	291	290	286	279	274	273	274	255	250		
	Edh at Noon		21	27	33	34	35	36	36	34	28	21	17	16		
(44) All-Sky Solar Radiation	RadAvg		422	797	1243	1614	1786	1862	1975	1623	1181	693	374	299		
	RadStd		32	53	91	109	156	135	94	130	96	90	52	29		

Historical Trends

		Heating		Cooling			Degree-Days					
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD50	HDD65	CDD50	CDD65		
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A
(47) Regional (19 neighbors)	N/A	N/A	N/A	N/A	N/A	-0.99	N/A	N/A	N/A	N/A	N/A	N/A

Nomenclature: See separate page